Counts, laws, denominators and API field names

Fix report g and information y.
Choose ν = s (supplied) or P (evaluation).
Weights: nonnegative and sum to one.

Original report frame

$M = \binom{n}{2} > 0, \quad x_{ij} = \mathbf{1}\{g_i < g_j\}$

$N(g) = \sum_{i \neq j} x_{ij}; \quad \text{raw disclosure} = N/M$

Weak direction error (≤)

$D_k(g, \Theta) =$

$\sum_{i \neq j} x_{ij} \mathbf{1}\{\theta_{ik} \leq \theta_{jk}\}$

Average under ν at fixed y ↓

$E_k^{\nu}(g) = \mathbb{E}_{\nu}[D_k \mid y]$

$E_k^{\nu}(g)/N(g), \quad N(g) > 0$

Strict reversal (<)

$\sum_{i \neq j} x_{ij} \mathbf{1}\{\theta_{ik} < \theta_{jk}\}$

Average under ν at fixed y ↓

$E_k^{b,\nu}(g)$ expected strict-error count

Two rates; different denominators:

$E_k^{b,\nu}(g)/N(g), \quad N(g) > 0$

$E_k^{b,\nu}(g)/Q_k^{\nu}(g), \quad Q_k^{\nu}(g) > 0$

Distinct opportunity (≠)

$\sum_{i \neq j} x_{ij} \mathbf{1}\{\theta_{ik} \neq \theta_{jk}\}$

Average under ν at fixed y ↓

$Q_k^{\nu}(g)$ expected distinct mass

Weighted distinct disclosure:

$Q_w^{\nu}(g)/T_w^{\nu}, \quad T_w^{\nu} > 0$

Weighting and total opportunity

$Q_w^{\nu}(g) = \sum_k w_k Q_k^{\nu}(g)$

$T_w^{\nu} = \sum_{i<j} \sum_k w_k d_{ij,k}^{\nu}$

$d_{ij,k}^{\nu} = \Pr_{\nu}(\theta_{ik} \neq \theta_{jk} \mid y)$

T covers all original unordered pairs.

Inspected tie_aware_v1 field names

API D ↔ $Q_w^s(g)$; API T ↔ T_w^s

Supplied-law aliases. Book D_k retains its realized weak wrong-count meaning.

Zero denominators stay undefined

$N = 0$: all disclosed counts are zero.
N-rates are NA; raw $N/M = 0$.
$Q_w^{\nu}/T_w^{\nu} = 0$ only if $T_w^{\nu} > 0$.
$Q_k^{\nu} = 0$: $E_k^{b,\nu}/Q_k^{\nu}$ is NA.
$T_w^{\nu} = 0$: Q_w^{ν}/T_w^{ν} is NA;
neutral risk and R_0^{ν} are zero.
Raw N can still be positive outside a guard-restricted family.

When whole reports or datasets vary,
let $A = \sum_k w_k D_k$.
E[A]/E[N] and E[A/N | N > 0]
are generally different summaries.
Name the law and positive denominator.
Different types can share numeric values.